

2010376103 CSE4261 Assignment-7

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1 Comparison among YOLO Versions: Object Detection Capability and Speed

[GitHub Code Link](#)

To compare object detection performance and speed, three Ultralytics models were selected: **YOLOv8-nano**, **YOLOv11-nano**, and **YOLOv12-nano**. A video of New York City street (without human faces) was processed by each model.

Figure 1 shows a representative frame and detections from each version. Quantitative results:

- YOLOv8n detected **4797** object instances, processing time: **3.56s**.
- YOLOv11n detected **4914** instances, processing time: **3.83s**.
- YOLOv12n detected **4334** instances, processing time: **5.18s**.

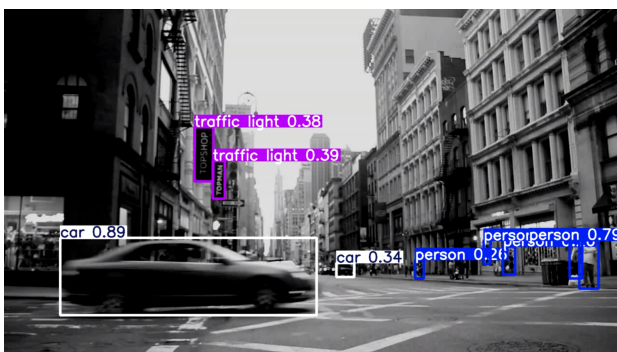
Observation: YOLOv11n detected slightly more objects, while YOLOv8n achieved the fastest inference.



(a) Original frame



(b) YOLOv8-nano



(c) YOLOv11-nano



(d) YOLOv12-nano

Figure 1: Comparison of detection performance among different YOLO versions

2 Fine-Tuning YOLOv8 on the WIDER FACE Dataset

[GitHub Code Link](#)

The **YOLOv8-nano** model was fine-tuned on the WIDER FACE dataset. The dataset was split into **training (90%)** and **validation (10%)** sets. Training was conducted for **10 epochs** with a batch size of 8.

Results: Validation mAP@50: **0.614**, Inference time: **3.0 ms/image**

Figure 2 shows the ground-truth labels and model predictions from validation batches.



(a) Validation batch-0 labels



(b) Validation batch-0 predictions



(c) Validation batch-1 labels



(d) Validation batch-1 predictions

Figure 2: Fine-tuned YOLOv8n on WIDER FACE validation results

3 Comparison Between Fine-Tuned YOLOv8n and GitHub YOLOv8-Face Model

Our fine-tuned YOLOv8n model was compared with the open-source [YOLOv8-Face](#) implementation.

- The GitHub YOLOv8-Face achieved higher validation AP (0.883) on WIDER FACE.
- Our model's AP was lower (0.614) due to shorter training (10 epochs).
- Inference speed was higher for our model (333 FPS on GTX 1080 vs. 169 FPS on RTX 4090).

Table 1: Comparison of YOLOv8-Face models

Metric	GitHub YOLOv8n-Face	Our YOLOv8n-Face
Avg. FPS	169 (RTX 4090)	333 (GTX 1080)
WIDER FACE Easy Val. AP	0.883	0.614

Observation: The GitHub version performs better in detection accuracy due to extended training and possible hyperparameter tuning, while our fine-tuned model demonstrates superior speed and efficiency.

4 Building a YOLOv1-Based Face Detector

[GitHub Code Link](#)

A custom **YOLOv1-based face detector** was implemented using PyTorch. The model architecture is illustrated in Figure 3. It was trained on the WIDER FACE dataset using a train-validation split (90:10), a learning rate of 1×10^{-3} , and 10 epochs.

Layer (type)	Output Shape	Param #
Conv2d-1	[-1, 64, 112, 112]	9,472
ReLU-2	[-1, 64, 112, 112]	0
MaxPool2d-3	[-1, 64, 56, 56]	0
Conv2d-4	[-1, 192, 56, 56]	110,784
ReLU-5	[-1, 192, 56, 56]	0
MaxPool2d-6	[-1, 192, 28, 28]	0
Conv2d-7	[-1, 128, 28, 28]	24,704
ReLU-8	[-1, 128, 28, 28]	0
Conv2d-9	[-1, 256, 28, 28]	295,168
ReLU-10	[-1, 256, 28, 28]	0
MaxPool2d-11	[-1, 256, 14, 14]	0
Conv2d-12	[-1, 256, 14, 14]	590,080
ReLU-13	[-1, 256, 14, 14]	0
Conv2d-14	[-1, 512, 14, 14]	1,180,160
ReLU-15	[-1, 512, 14, 14]	0
MaxPool2d-16	[-1, 512, 7, 7]	0
Conv2d-17	[-1, 512, 7, 7]	2,359,808
ReLU-18	[-1, 512, 7, 7]	0
Conv2d-19	[-1, 1024, 7, 7]	4,719,616
ReLU-20	[-1, 1024, 7, 7]	0
Conv2d-21	[-1, 1024, 7, 7]	9,438,208
ReLU-22	[-1, 1024, 7, 7]	0
Conv2d-23	[-1, 1024, 7, 7]	9,438,208
ReLU-24	[-1, 1024, 7, 7]	0
Flatten-25	[-1, 50176]	0
Linear-26	[-1, 4096]	205,524,992
ReLU-27	[-1, 4096]	0
Dropout-28	[-1, 4096]	0
Linear-29	[-1, 539]	2,208,283
Total params: 235,899,483		
Trainable params: 235,899,483		
Non-trainable params: 0		
Input size (MB): 0.57		
Forward/backward pass size (MB): 34.74		
Params size (MB): 899.89		
Estimated Total Size (MB): 935.20		

Figure 3: YOLOv1-based face detector architecture

5 Summary

- YOLOv8n was fastest among YOLOv8/11/12 variants.
- YOLOv11n showed slightly higher detection coverage.
- Fine-tuned YOLOv8n achieved $\text{mAP}_{50} = 0.614$ and very high FPS.
- GitHub YOLOv8-Face achieved higher accuracy ($\text{AP} = 0.883$).
- YOLOv1 model successfully trained but remains less capable than YOLOv8-based detectors.